



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

APACHE KAFKA PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Messaging & Streaming

TOTAL: 10 QUESTIONS

Q1 What is Apache Kafka and what problem in Messaging & Streaming does it solve? What Model

Q6 How does Apache Kafka handle reliability and high availability? Observability

Q2 What are the core components or concepts in Apache Kafka? Architecture

Q7 What observability does Apache Kafka provide out of the box? Observability

Q3 How does Apache Kafka compare to its main alternatives? Alternatives

Q8 What are common performance bottlenecks in Apache Kafka? Performance

Q4 How do you get started with Apache Kafka for a new project? Getting Started

Q9 What security best practices apply when running Apache Kafka? Security

Q5 What configuration options most affect Apache Kafka's behavior in production? Configuration

Q10 What mistakes do teams commonly make when adopting Apache Kafka? Mistakes



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Apache Kafka is a tool or platform

Mitigation

Apache Kafka is a tool or platform that automates or simplifies a specific concern in the Messaging & Streaming domain, reducing manual effort and operational complexity for engineering teams.

A6 Apache Kafka provides HA through leader election, Observability

Apache Kafka provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Apache Kafka has key building blocks that

Primitives

Apache Kafka has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Apache Kafka exposes metrics (Prometheus endpoint or Information

Apache Kafka exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Apache Kafka trades off simplicity against feature

Hardening

Apache Kafka trades off simplicity against feature richness differently than its alternatives. Choose Apache Kafka when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Apache Kafka via its package manager

Gotchas

Install Apache Kafka via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Apache Kafka with the principle of

Assessment

Run Apache Kafka with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency, Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.